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PUSL3190 Computing Individual Project

Abstract

MozziTrack

**Development of a Centralized Dengue Prevention
Platform for Public Health Inspectors in Sri Lanka**

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Abstract

This project focuses on developing a mobile-responsive web platform to improve dengue prevention and control efforts in Sri Lanka. Recognizing the increasing threat of dengue fever, especially in urban areas with poor waste management and rapid population growth, it is clear that the current paper-based inspection methods used by health officers are slow, prone to delays, and lack a central system to store or track data. These limitations make it difficult to respond to outbreaks quickly and effectively. To address this, a digital platform was developed to replace the manual process.

The platform enables Public Health Inspectors (PHIs) to digitally record site inspections, update complaint statuses, and schedule follow-ups efficiently. Citizens can track the progress of complaints related to their own properties and receive real-time updates on dengue outbreaks. Through a community chat feature, they can also report concerns and share dengue prevention tips. Administrators and the National Dengue Control Unit (NDCU) are provided with dynamic dashboards that display outbreak trends and risk zones, helping them make better decisions based on real-time data.

The platform was built using React.js, Node.js, MySQL, and Power BI to ensure fast access, organized workflows, and effective data analysis. Although the web application has been successfully developed and tested, a dedicated mobile application is planned for future development to further enhance accessibility. This project demonstrates how digital tools can improve public health systems by making them faster, more efficient, and more connected with the community.

Keywords:

Dengue Prevention, Digital Health System, Public Health Inspectors, Community Engagement, Disease Control, Sri Lanka